

Research article

Towards targeted cancer therapy: Synthesis, characterization, and biological activity of a new Cu(II)-ibuprofen-2,2'-dipyridylamine metal complex

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A B S T R A C T

This work reports the synthesis of a copper metal complex with the nonsteroidal anti-inflammatory drug (NSAID) ibuprofen, and 2,2'-dipyridylamine employing microwave-assisted synthesis (MWAS). To the best of authors knowledge, this is the first study reporting a NSAID-based complex achieved through MWAS. The coordination compound was characterised by elemental analysis, Fourier transform infrared spectroscopy, thermogravimetry, and ultraviolet–visible spectrophotometry. Additionally, the crystal structure of the copper metal complex was elucidated using single-crystal X-ray diffraction with synchrotron radiation. The compound's interaction with the biomolecules bovine serum albumin (BSA) and calf-thymus DNA (CT-DNA), was assessed through UV–Vis, circular dichroism, and fluorescence spectroscopy. Our findings demonstrate that the metal complex effectively binds to BSA, causing a reduction in its intrinsic fluorescence and α -helical content, and shows a capacity for intercalation between CT-DNA base pairs. Finally, the copper compound exhibited promising *in vitro* antitumoral activities against human breast cancer cell lines (MCF-7 and MDA-MB-231), as evaluated by MTT (3-[4,5-dimethylthiazol-2-yl]-2,5 diphenyl tetrazolium bromide) assay (although a similar cytotoxic effect against a non-tumoral epithelial cancer cell line, MCF-12A, was found), and increased oxidative stress levels as assessed by the TBARS (thiobarbituric acid reactive substances) assay and by evaluating glutathione levels. The results suggest that the metal complex promotes lipid peroxidation by increasing oxidative stress levels, leading to a reduction in viability of the two breast cancer cell lines.

1. Introduction

The exploration of coordination compounds of transition metals and nonsteroidal anti-inflammatory drugs (NSAIDs) originated from the development of a copper(II)-acetylsalicylic acid (aspirin) metal complex [1]. Remarkably, this compound exhibited enhanced anti-inflammatory efficacy and reduced side effects compared to aspirin alone [2]. NSAIDs constitute a broad class of commercially available drugs featuring diverse structural and functional attributes. Most NSAIDs are weak organic acids containing carboxylic functional groups that can deprotonate under physiological pH [3] thereby offering versatile coordination modes with metal ions [4, 5]. Combining NSAIDs with transition metals such as copper, zinc and platinum in coordination complexes can provide synergistic effects, enhancing both anti-inflammatory and other therapeutic properties (e.g., antimicrobial or anticancer effects) while reducing

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adverse effects, making these novel inorganic compounds intriguing candidates for pharmaceutical applications [6,7]. The properties of such NSAID-metal complexes can be fine-tuned, allowing for adaptability in treatments. Their features, such as oxidation states, charges, and spatial configurations, are highly modifiable [8]. This versatility has led to their application in various therapeutic areas, including cancer treatment, microbial inhibition, and ulcer therapy [9]. Recently, NSAIDs-metal complexes were found to be one of the most promising classes of anti-cancer stem cells agents [10,11]. Despite significant advancements in the synthesis of NSAID-metal complexes over recent decades (mainly employing one-pot synthesis methods [12]) the challenge of obtaining high quality crystals remains [13], often requiring time-consuming and variable crystallization conditions. Traditional synthesis involves a mixture of NSAID, a metal salt, and sometimes an N-donor molecule in volatile solvents like methanol or ethanol [14,15] followed by a lengthy crystallization process [16,17]. Microwave-assisted synthesis (MWAS) emerges as a highly controlled alternative, offering benefits like reduced crystallization time, uniform particle size, and phase selectivity. Despite its advantages and broad application in both organic [18] and inorganic chemistry [19] to the best of our knowledge MWAS has not yet been applied in synthesizing NSAID-based metal complexes.

In this study, we report the innovative synthesis and detailed characterization of a novel Cu(II) metal complex with ibuprofen (ibu) and 2,2'-dipyridylamine (dpaH) using a one-pot MWAS approach. Despite its long history, new metal complexes of dpaH continue to be discovered, making this ligand a gift to coordination chemists as highlighted by Kumar and co-authors [20]. While dpaH has received less attention compared to other ligands in the bipyridine family, it has been used in the synthesis of metal complexes with anticancer properties [21,22]. Therefore, it is plausible that combining copper, ibu, and dpaH could result in a metal complex with potential anticancer activity. The resulting compound $[\text{Cu}(\text{Ibu})(\text{dpaH})\text{Cl}]$ was fully analysed via elemental analysis, Fourier transform infrared spectroscopy (FTIR), thermogravimetric analysis (TGA), and UV-Vis spectroscopy. We also determined its crystal structure through X-ray crystallography. Furthermore, interaction studies with bovine serum albumin (BSA) and calf-thymus DNA (CT-DNA) were conducted by UV-Vis, circular dichroism (CD), and fluorescence spectroscopy. Finally, its antitumoral efficacy was assessed through *in vitro* assays against human breast cancer and non-cancer cell lines (MCF-7, MDA-MB-231 and MCF-12A), solidifying its potential as a candidate for anticancer applications.

2. Results and discussion

2.1. Synthetic strategy

In this work, the MWAS method allowed the synthesis of complex 1 after only 10 min of reaction time, albeit with some unknown crystalline impurity (denoted as an asterisk in Fig. S1, orange diffractogram), which was absent after 60 min. Extending the reaction time further had no impact on the overall purity of the material. Although complex 1 can be synthesized through slow evaporation at room temperature, this method was time-consuming (taking up to one week) and failed to eliminate a minor impurity (denoted as an asterisk in Fig. S1, red diffractogram). With the MWAS approach we were able to considerably reduce the reaction time, without significant change in the overall yield and obtaining a pure-phase material. Regarding crystal morphology, both approaches lead to the formation of large crystals with a plate-like morphology (Fig. S2). The slow evaporation allowed the formation of slightly larger crystals, with sizes between 50 and 200 μm while the MWAS approach originated crystals with sizes ranging from 25 to 75 μm .

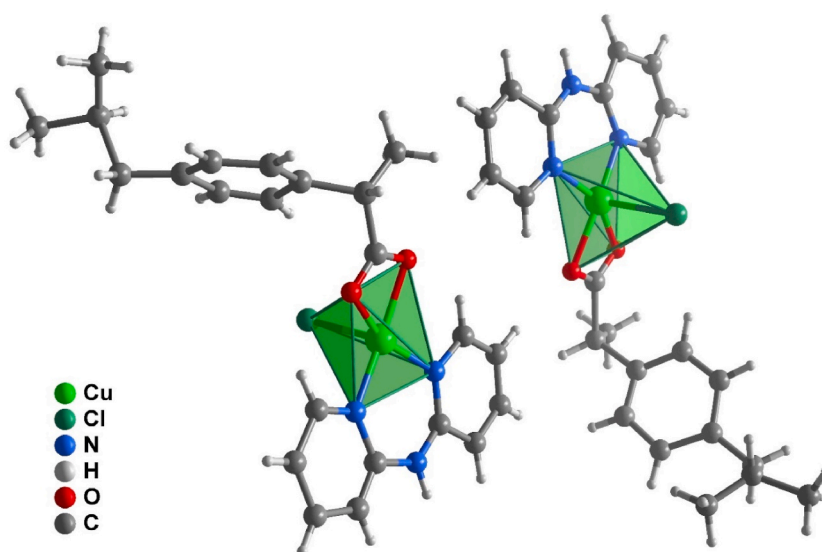


Fig. 1. Asymmetric unit of complex 1, $[\text{Cu}_2(\text{Ibu})_2(\text{dpaH})_2\text{Cl}_2]$, emphasizing the square pyramidal coordination geometry of the copper cation.

2.1.1. Crystal description

The structure of complex 1 was unveiled from single-crystal X-ray diffraction analysis using synchrotron radiation (BM01 - Swiss-Norwegian beamline at the ESRF). Compound 1 crystallizes in the monoclinic non-centrosymmetric space group $P2_1$ being formulated as $[\text{Cu}(\text{Ibu})(\text{dpaH})\text{Cl}]$. The asymmetric unit consists of two identical units, each incorporating a molecule of ibu, a molecule of dpaH and a chloride anion, all of which are coordinated to a central Cu^{2+} (Fig. 1). The Cu^{2+} metal cation is pentacoordinated, presenting a $\{\text{CuO}_2\text{N}_2\text{Cl}\}$ coordination sphere that resembles a slightly distorted square pyramid. The dpaH ligands are coordinated to Cu^{2+} in a bidentate manner, via N,N -chelation. These ligands assume a near-planar orientation, with the angle between their two pyridine rings measuring approximately 3.8° . Conversely, the ibu molecules coordinate to the central metal ion, also in a bidentate manner, but through a O,O -chelate originating from their carboxylic groups. The crystal packing (Fig. S3 in the ESI) is governed primarily by weak $\text{C-H}\cdots\text{C}$ hydrogen bonds, mainly involving the ibuprofen components, and by strong π - π interactions. Because of the planar nature adopted by dpaH, the complexes stack atop each other along the a -axis of the unit cell. This stacking is facilitated by a 180° -degree rotation along the same axis (Fig. S4 in the ESI), resulting in multiple π - π interactions with distances ranging from $3.543(10)$ to $3.649(10)$ Å. Additional structural stability is achieved by intermolecular interactions arising from hydrogen bonds between the chloride anion coordinated to Cu^{2+} and the protonated nitrogen atom of the 2,2'-bipyridylaminium moieties. These hydrogen bonds exhibit distances in the range of $3.241(19)$ to $3.10(2)$ Å and interaction angles ($\text{N-H}\cdots\text{Cl}$) ranging from 155° to 167° . Details of the crystallographic data are summarized in Tables S1 and S2 (in the ESI).

2.2. FTIR and thermogravimetric analysis

FTIR analysis was performed to explore the presence of ligands and their coordination modes in the complex (Fig. S5 in the ESI). The presence of ibu is evidenced by the bands in the regions $3100\text{--}2900$, $2980\text{--}2850$, and $1650\text{--}1600\text{ cm}^{-1}$ assigned to the $\text{C}(\text{sp}^2)\text{--H}$, C--H (aliphatic) and $\text{C}=\text{C}$ stretching vibration modes, respectively [23,24]. Coordination of the NSAID to the metal centre is witnessed by the absence of the strong band characteristic of the free acid group at 1710 cm^{-1} and the presence of two bands at 1536 and 1437 cm^{-1} , attributed to the $\nu_{\text{asym}}(\text{COO}^-)$ and $\nu_{\text{sym}}(\text{COO}^-)$ stretching vibration modes of the carboxylate group. The difference $\Delta = \nu_{\text{asym}}(\text{COO}^-) - \nu_{\text{sym}}(\text{COO}^-)$ gives information on the coordination mode of the carboxylate group [24,25], lower than 200 cm^{-1} is bidentate, higher is monodentate. For complex 1 is $\Delta = 150\text{ cm}^{-1}$, indicating a bidentate binding mode, in accord with the crystal structure. In addition, the $\nu(\text{NH})$ band at 3250 cm^{-1} shows that the dpaH NH group is not involved in metal coordination [26]. Both, the pyridyl ring stretching bands at 3095 and 1500 cm^{-1} , and the out of plane $\tau(\text{C--H})$ vibrations of the N donor ligand at $3030\text{--}3050\text{ cm}^{-1}$ and 775 cm^{-1} , confirm the presence in the complex of the nitrogen donor ligand [26]. Lower intensity bands like $\gamma(\text{M--N})$ and $\gamma(\text{M--O})$ are also observed at 487 and 565 cm^{-1} , proving the coordination to dpaH and ibu, respectively [25,26]. Thermogravimetry shows a three-stage decomposition (Fig. S6 in the ESI). The first step between 25 and 340°C is attributed to the loss of one molecule of dpaH (observed weight loss 34% , calculated 36%). The second step between 340 and 720°C is ascribed to the degradation of one chloride ion (most probably in the form of hydrochloric acid) and one molecule of ibu (observed weight loss 52% , calculated 51%). Further heating results in the final CuO residue (confirmed by powder XRD, Fig. S7 in the ESI).

2.3. UV-Vis spectrophotometry

The absorption spectra of the free ligands and complex 1 in ethanol are shown in Fig. S8 (in the ESI). The bands attributed to π bonds in free dipyriddy molecules are observed at 265 and 315 nm . The red shift of these bands can take place when the ligand reacts with a metal ion to form a chelate ring [27]. Free ibu shows absorption bands at 222 and 270 nm assigned to π - π^* transitions [28]. Both characteristic dpaH and ibu bands are observed in the spectrum of the compound and present a red shift (bathochromic effect) highlighting their coordination to the metal centre. It is also possible to observe a hyperchromic effect due to metal-to-ligand charge transfer transitions observed for the complex and that is attributed to coordination with ligands. Analysis of complex 1 at higher concentration revealed the presence of a low intensity band (at around 675 nm) attributed to d-d transitions, typical for distorted square pyramidal geometry [29] (Fig. S9 in the ESI). This finding is in agreement with the electronic structure of Cu^{2+} with a d^9 electronic configuration [30].

2.4. BSA interaction studies

2.4.1. Fluorescence quenching and binding constant

The binding affinity between bioactive molecules or compounds and serum proteins like albumin in the bloodstream is a critical factor that influences their distribution, free concentration levels, metabolic rates, and potential toxicity. Hence, gaining insights into the interaction mechanisms with serum albumins is crucial for understanding their pharmacokinetics and pharmacodynamics [31–33]. Among serum albumins, BSA is the most comprehensively studied due to its structural similarity to Human Serum Albumin [31]. In our study, a solution of BSA was titrated at room temperature using incremental concentrations of Complex 1. Fig. 2 illustrates the effect of this complex on BSA's fluorescence emission spectrum. Notably, we observed a consistent decrease in BSA's fluorescence intensity - up to 59.6% - as the concentration of complex 1 increased, corroborating the interaction between the two substances. The maximum fluorescence emission was attained at a wavelength of 342.5 nm , accompanied by a minor red shift in the spectrum. This fact suggests an increase in the hydrophobicity of the microenvironment surrounding the tryptophan residues as a consequence of complex binding [34]. The same behaviour has been previously observed in the interaction between other Cu-NSAIDs and BSA [25,35–37]. The Stern-Volmer plot obtained for complex 1 is linear and allowed to calculate the value of $K_{\text{sv}} = 1.54 (\pm 0.05) \times 10^4\text{ M}^{-1}$. The linear plot

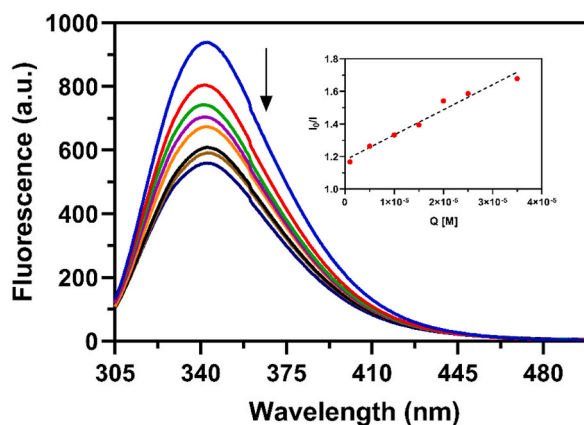


Fig. 2. Fluorescence emission spectra of BSA in the absence and presence of complex 1 at $\lambda_{ex} = 295$ nm and $[BSA] = 5 \mu M$. Arrow indicates the change upon increasing the complex concentration ($R = 0.2$; $R = 1$; $R = 2$; $R = 3$; $R = 4$; $R = 5$; $R = 6$; $R = 7$; $R = \text{Protein to complex ratio}$). The plot F_0/F versus $[Q]$ according with the Stern-Volmer equation is depicted in the inset; $[Q]$ = concentration of quencher (M).

obtained (Fig. 2 – inset) indicates that only one type of quenching mechanism is involved and according to the K_{sv} value the interaction between complex 1 and BSA is moderately strong [34].

2.4.2. Circular dichroism spectroscopy

When drugs or other bioactive compounds bind to proteins, the intermolecular forces that stabilize the protein's secondary and tertiary structures can be affected, often leading to conformational changes in the protein. CD spectroscopy is a valuable analytical technique for exploring the specific binding interactions between small substrates and chiral macromolecules helix [38]. Accordingly, we employed CD spectroscopy to monitor potential conformational changes in BSA upon its binding with the metal complex. Native BSA exhibited two negative bands at 208 and 222 nm which are typical of the α -helical structure of proteins. These negative peaks are the contribution from the $\pi \rightarrow \pi^*$ (208 nm) and $n \rightarrow \pi^*$ (222 nm) transitions of the peptides in the α -helix [39,40]. As depicted in Fig. 3, complex 1 was able to reduce the two negative bands of the native BSA along with a slight red shift. This clearly indicates decrease in the α -helical content of the protein. We have also determined the α -helical content of free and bound BSA with complex 1, calculated from MRE values at 208 nm using equation (2) as discussed in section 2.5. According to quantitative analysis, native $[BSA] = 0.25 \mu M$ has ca. 37 % of α -helical content and complex 1 was able to reduce this percentage in about 6 and 7 % for $R = 0.5$ and $R = 1.0$, respectively.

The observed reduction in the α -helical content implies that the metal complex induces a slight unfolding of the polypeptides within BSA. Given that a protein's secondary structure largely reflects local conformations within its polypeptide chains and is predominantly stabilized by hydrogen bonds, our findings suggest that complex 1 perturbs these hydrogen bonding interactions [41,42]. Such conformational alterations could result in greater exposure of hydrophobic regions that were previously sequestered within the protein's structure [42]. It's worth noting that similar conformational changes have been reported for other Cu(II) complexes [41,43,44]. However, according to literature a copper-ibuprofenate metal complex produce a contrasting effect, leading to an increase in BSA's α -helical content [45].

2.4.3. UV-vis spectroscopy

The UV-Vis spectrum of BSA exhibit two absorption bands at 208 and 280 nm. The first band corresponds to the $\pi \rightarrow \pi^*$ transitions of BSA's characteristic polypeptide backbone, while the second one is associated with the polarity of the microenvironment around the

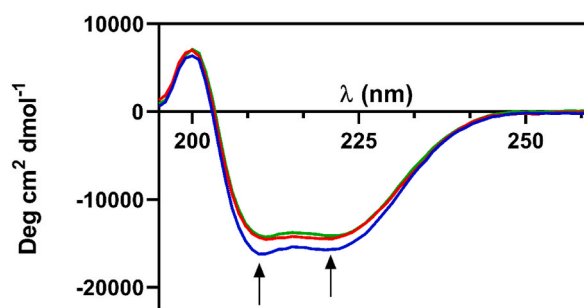


Fig. 3. CD spectra of BSA in the presence and absence of complex 1. Arrow indicates the change upon increasing the complex concentration. ($[BSA] = 0.25 \mu M$; $R = 0.5$; $R = 1$).

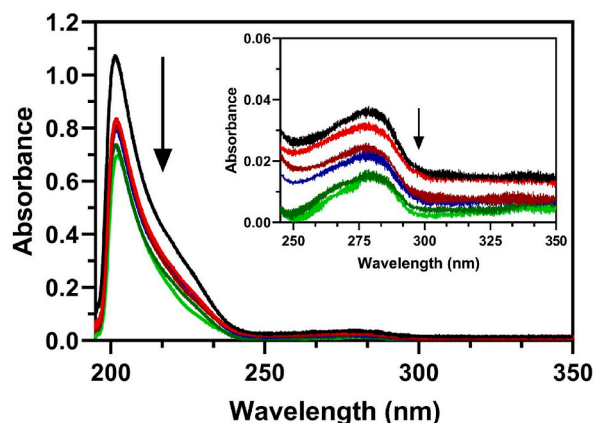


Fig. 4. Absorption spectra of BSA in the absence (a) and presence of different concentrations of complex 1. [BSA] = 0.47 μ M; R = 0.21; R = 2.13; R = 10.6; R = 21.3; R = 53.2.

tryptophan and tyrosine residues (aromatic ring of amino acids) [46]. This method is also useful for determining the potential quenching mechanism of metal complexes. In cases of dynamic quenching, the interaction between the quencher and the protein does not cause changes in the BSA absorption spectrum. However, if static quenching occurs, the spectrum will be significantly altered [47, 48].

In Fig. 4, we can see that the 208 nm band decreases with the addition of the complex, indicating an interference in the protein's secondary structure (peptide bonds). The changes at the 280 nm peak are less marked, suggesting that the environment around the aromatic rings of the amino acids has been slightly modified. These results are indicative of a static interaction between BSA and complex 1 and confirm fluorescence findings, that only one type of quenching mechanism is involved.

2.5. CT-DNA binding studies

2.5.1. UV-vis absorption spectroscopic studies

UV-Vis absorption spectroscopy is one of the most valuable techniques to study the binding interactions between drugs and CT-DNA. These interactions are discernible through changes in the absorption spectra before and after the introduction of metal complexes [49–51]. According to existing literature, the UV spectrum of free CT-DNA displays a prominent absorption band at 256 nm, attributable to $\pi \rightarrow \pi^*$ intraligand transitions [52]. Hyperchromic and hypochromic effects are spectral hallmarks associated with the double-helical structure of DNA [53] and the degree of hypochromism generally serves as an indicator of the strength of intercalative drug binding [54]. In the case of intercalation, a complex binds to CT-DNA, leading to robust π - π stacking interactions between the planar aromatic rings of a coordinated ligand and the DNA base pairs. Such interactions typically manifest as hypochromic and a redshift in the spectrum of either the ligand or charge transfer bands [55]. Our study presents the UV-Vis absorption spectra of CT-DNA both in the absence and presence of complex 1 (Fig. 5). We observed a hypochromic effect, characterized by 17 % reduction in absorbance, along with a bathochromic shift. These spectral features are commonly indicative of molecules or metal complexes capable of intercalating with DNA [56]. Considering the crystallographic data, the planar nature of the dpaH ligand in complex 1

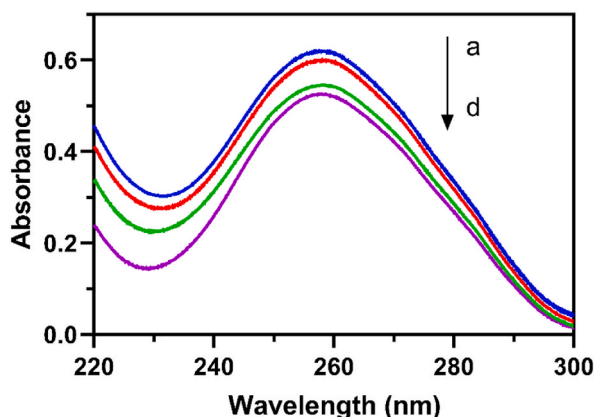


Fig. 5. Absorption spectra of CT-DNA in the absence (a) and presence of different concentrations of complex 1. ([CT-DNA] = 0.5 μ M; [complex 1] = b-5 μ M; c-15 μ M; d-30 μ M).

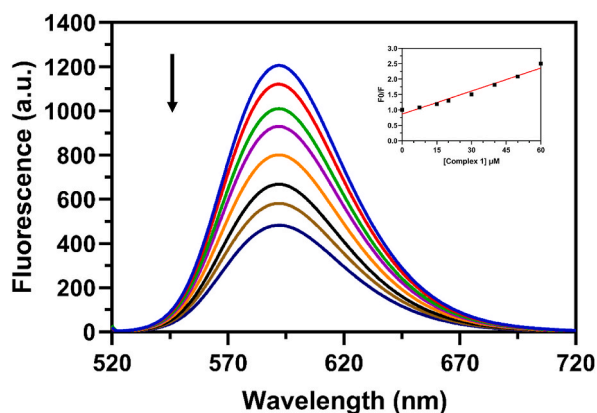


Fig. 6. Effect of the addition of the metal complex on the fluorescence intensity of the DNA-EB system, [complex 1] = 5–60 μM . Arrow shows the effect of complex 1 on the adduct fluorescence.

appears conducive to π - π interactions with the base pairs of CT-DNA. Thus, the UV-Vis spectroscopic analysis suggests the complex's ability to engage with DNA through intercalative binding.

2.5.2. Competitive DNA binding studies

Ethidium Bromide (EB) is a highly sensitive fluorescence probe commonly used for DNA binding studies. Its fluorescence intensity significantly increases upon intercalating into CT-DNA. This heightened fluorescence is attributed to the strong intercalation of EB's planar phenanthridine ring between DNA base pairs [57–59]. When a metal complex intercalates into DNA, it reduces the available binding sites that EB can occupy. This leads to a subsequent decrease in fluorescence intensity—also referred to as "quenching"—in the EB-DNA system [59–61]. Our competitive binding study results are displayed in Fig. 6. Upon the addition of complex 1 to the EB-DNA system, a noticeable quenching of fluorescence intensity was observed. This outcome suggests that the compound successfully competed with EB for DNA intercalation sites and displaced it from its original position [61]. These findings are in agreement with other reports [62–64] that indicate intercalation as the most possible interaction mode of Cu(II)-NSAIDs metal complexes with CT-DNA. Data also corroborate the results from the absorption spectroscopic study, confirming the intercalative binding capability of the metal complex.

2.5.3. Circular dichroism spectroscopy

The CD spectrum of CT-DNA exhibit two typical bands, a positive one at 275 nm and a negative at 245 nm that are characteristic of the B-DNA form in solution. The positive band is due to the base stacking interactions and the negative is due to right-handed helicity of DNA [65]. Different interactions between small molecules and the DNA show different perturbations in the CD spectrum of the biomolecule, for example, simple groove binding or electrostatic interactions show very small perturbations or no perturbation at all on the base stacking and helicity bands. On the other hand, intercalation cause significant changes in both bands [66].

As shown in Fig. 7, the addition of the copper complex to the DNA solution leads to alterations in both bands, indicating that the complex disrupts DNA base stacking and interfered with right-handed helicity. This suggests that the coordination compound interacts with the DNA double strands through intercalation. These findings are consistent with the results previously obtained from UV-Vis and fluorescence studies.

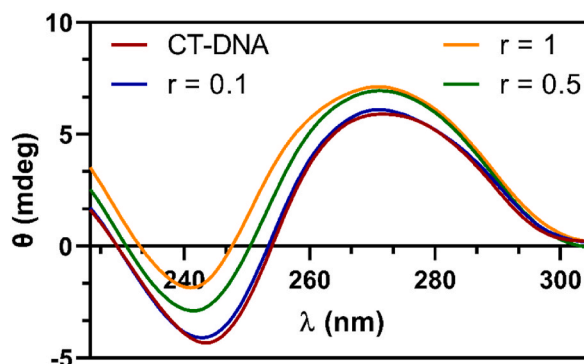


Fig. 7. CD spectra of CT-DNA in the presence and absence of complex 1. ([CT-DNA] = 50 μM ; R = 0.1; R = 0.5; R = 1).

2.6. *In vitro* biological assays

2.6.1. MTT assay

Cell viability studies were performed against two human tumoral breast cell lines with distinct characteristics, MCF-7 and MDA-MB-231. The MCF-7 cancer cell line is ER-positive, being considered a poorly aggressive and non-invasive cell line with a low metastatic potential [67]. On the other hand, the MDA-MB-231 cell line is a triple-negative breast cancer cell line, being very aggressive and associated to a highly invasive behaviour [68,69]. The MTT studies revealed a concentration-dependent cytotoxic effect for the two cell lines under study, with a similar potency, after a 24h exposure to complex 1. Indeed, for the highest tested concentration (25 μ M), the coordination compound reduced the cellular viability by 30 % in both cell lines (Fig. 8a), with IC_{25} values of 20.7 (15.5–27.6) μ M and 18.7 (11.8–30.4) μ M, respectively – Table S3 in the ESI. The potency of complex 1 is in the same range of the potency of the commercially available antitumor agent cisplatin (IC_{50} in MCF-7: 28–93 μ M and in MDA-MB-231: 39–82 μ M) [70,71].

Similar to the present observations, previous studies have shown that copper-NSAIDs present anti-proliferative properties. A series of Cu(II) metal complexes with derivatives of salicylic acid (NSAID) with and without N-donor ligands (namely phenanthroline, phen) were tested for their *in vitro* cytotoxicity, using standard MTT assay, against breast (MCF-7), prostate (DU145), colon (HT29) and ovarian (SK-OV-3) cancer cell lines. Complexes containing bis-chelate (salicylate-phen) exhibit rapid low-micromolar cytotoxicity over 24h against all the cancer cells ($30 < IC_{50} < 50 \mu$ M). In contrast, simple complexes (not containing the chelating phen ligand but only salicylic acid derivatives) exhibited moderate activity against the four cell lines ($IC_{50} > 100 \mu$ M) [72]. In another study, Piroš et al. reported the antiproliferative properties of three copper clonixinates with methylpyridines (α -, β - and γ -picoline) ligands, [Cu (clon) $_2$ (α -pic) $_2$]; [Cu(clon) $_2$ (β -pic) $_2$]; and [Cu(clon) $_2$ (γ -pic) $_2$], against three human cancer cell lines including MCF-7. After 72h of incubation, all three complexes exhibit significant cytotoxicity ($29 < IC_{50} < 75 \mu$ M) against MCF-7 cancer cell line [73]. Our cytotoxic results are in line with both studies and highlight the anticancer potential of complex 1 against the ER-positive breast cancer cell line, MCF-7. However, regarding MDA-MB-231 cell line, our findings are difficult to compare with other reports because most of the cytotoxic studies did not evaluate the antiproliferative activity of Cu(II)-NSAIDs metal complexes against this cell line [74,75]. This gap is specially surprising taking in consideration that MDA-MB-231 cell line is a triple-negative breast cancer cell line, being very aggressive and associated to a highly invasive behaviour. Of note, complex 1 exhibits no significant selectivity against cancer cells compared to the normal MCF-12A cell line (Fig. 8a) (IC_{25} of 12.7 (4.4–19.3) μ M – Table S3 in the ESI). This property of complex 1 was

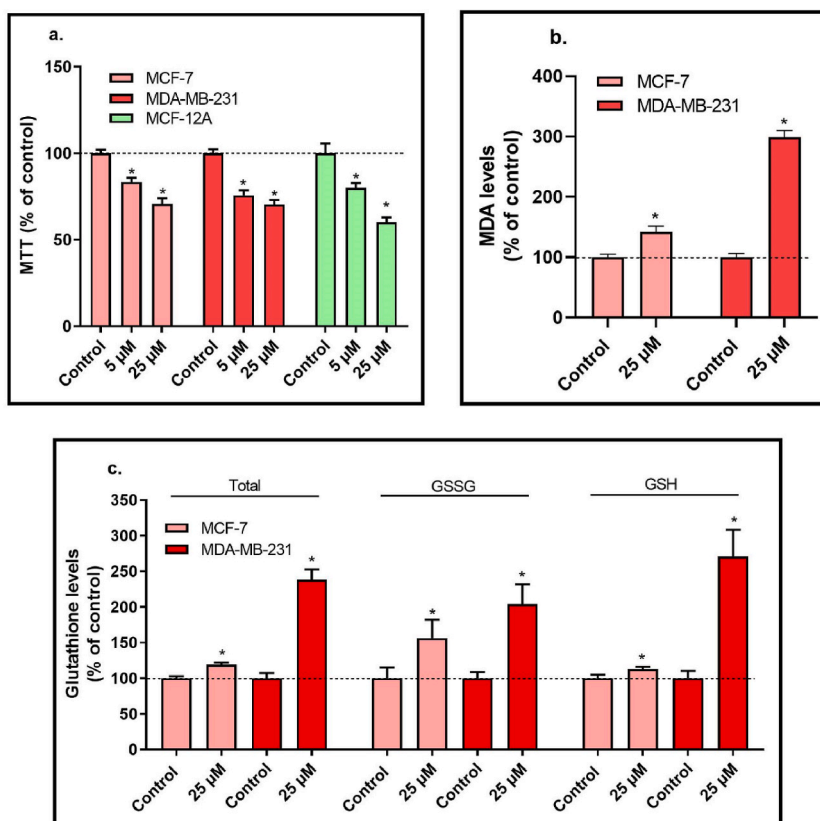


Fig. 8. Cell viability (a), lipid peroxidation (MDA) levels (b) and total, oxidized (GSSG) and reduced (GSH) glutathione levels (c) after treatment with complex 1 for 24h (n = 6). * $p < 0,05$ vs control (t-test).

also verified with other copper-NSAIDs complexes [73], highlighting the need for further studies aiming at developing copper-NSAIDs complexes with a higher selectivity for cancer cells.

2.6.2. MDA levels

Lipid peroxidation (LPO) is a natural process in the cellular system and one of the most dramatic consequences inflicted by reactive oxygen species (ROS) generation upon biological membranes structure and function. It is also known that lipid hydroperoxides and oxygenated products of lipid degradation, as well as initiators (i.e., ROS), can contribute to cell proliferation, signal transduction cascade, differentiation, and apoptosis [76,77]. Consequently, oxidative stress caused by increasing amounts of ROS may play an important part in the development of cancer and on its treatment with chemotherapeutic drugs [78–80]. Cells are continuously producing ROS as a consequence of their metabolic processes; however, the balance between the prooxidant and antioxidant levels keeps the biological equilibrium of the redox states [81]. Most chemotherapeutic drugs (for example, cisplatin and doxorubicin) increase ROS intracellular levels [81]. Therefore, we have quantified LPO levels (MDA levels in %) in MCF-7 and MDA-MB-231 cells exposed to complex 1, using the TBARS method (Fig. 8b). The results showed that MDA levels increased significantly, after a 24h exposure to complex 1, in both cell lines, when compared with control. The observation that this increase was especially evident in the MDA-MB-231 cell line may be related to the inherent differences in oxidative stress/reactive oxygen species levels in MCF-7 (less aggressive and invasive potential) and MDA-MB-231 (more aggressive and invasive potential) cells [82].

Therefore, the TBARS assay supports the conclusion that complex 1 promotes lipid peroxidation by increasing oxidative stress levels, leading to a reduction in cell viability; however, we cannot exclude that other mechanisms may contribute to the observed cytotoxic effect [83]. The results correlated well with the MTT study and are similar to the ones reported by Hussain et al. [84] regarding the change in lipid peroxidation levels of MCF-7 cells (evaluated using the TBARS method) exposed to Cu(II)-benzimidazole complexes. To the best of authors knowledge there are no studies reporting LPO levels for cells exposed to NSAIDs-based metal complexes using the TBARS method. However, Boodram and co-authors reported a series of Cu(II)-phenanthroline complexes bearing the NSAID, indomethacin, that were able to selectively killed breast cancer stem cells by elevating intracellular reactive oxygen species (ROS) levels and inhibiting cyclooxygenase-2 (revealed by mechanistic studies) [85].

2.6.3. Total, oxidized and reduced glutathione levels

The consequences of cell exposure to complex 1 on the cellular antioxidant defenses were evaluated by assessing its effect on glutathione, a major intracellular antioxidant. The exposure to complex 1 caused a marked increase in total, oxidized and reduced glutathione levels, suggesting that a compensatory antioxidant mechanism/upregulation of glutathione system developed in order to overcome higher oxidative stress levels (Fig. 8c). Again, the difference in the magnitude of the effect between MCF-7 and MDA-MB-231 cells may well be related to inherent differences in oxidative stress/reactive oxygen species levels in these two breast cancer cell lines. Lastly, it is important to highlight that the biological effects of complex 1 are significantly different from those of ibuprofen, which showed no cytotoxicity, did not enhance lipid peroxidation, and had only a minimal impact on glutathione levels (as shown in Fig. S10 of the ESI).

3. Conclusions

In this study, we present a highly efficient MWAS synthetic method, described here for the first time, for the synthesis of complex 1, a copper-based coordination compound featuring ibuprofen and 2,2'-dipyridylamine as ligands. The metal complex exhibited a mononuclear non-centrosymmetric structure. Spectroscopic techniques, including fluorescence, circular dichroism, and UV-Vis revealed that the compound can effectively interact with BSA. This interaction leads to discernible changes in the protein's secondary structure, characterized by a reduction in α -helical content and alterations in the microenvironment surrounding the tryptophan residues. The CT-DNA binding studies indicated that intercalation is the mechanism through which the compound interacts with this biomolecule. Biologically, the metal complex demonstrated cytotoxic effects against two human breast cancer cell lines, MCF-7 and MDA-MB-231, although a similar cytotoxic effect against a non-tumoral epithelial cancer cell line, MCF-12A, was found. Furthermore, the compound disrupted the redox balance in both cancer cell lines, which may contribute to a decrease in cell viability. The biological effects of complex 1 differ significantly from ibuprofen, which showed no cytotoxicity, no increase in lipid peroxidation, and minimal impact on glutathione levels. In conclusion, our findings underscore the therapeutic potential of this novel copper NSAID-based complex as an anticancer agent, owing to its capability to interact with key biological targets and disrupt cellular redox homeostasis. Future studies could explore other mechanisms of action and assess the complex's *in vivo* efficacy.

CRediT authorship contribution statement

Luís P.G. Monteiro: Writing – original draft, Investigation. **A. Gomes:** Investigation. **C. Silva:** Validation, Investigation. **Ricardo F. Mendes:** Software, Investigation, Data curation. **Filipe A. Almeida Paz:** Validation, Software, Resources. **Dmitry Chernyshov:** Resources, Data curation. **J. Rocha:** Writing – review & editing, Resources, Project administration. **F. Martel:** Validation, Resources. **T.M. Santos:** Writing – review & editing, Supervision, Conceptualization. **B.J.M. Leite Ferreira:** Writing – review & editing, Supervision, Conceptualization.

Data and code availability statement

Data included in article/supplementary material is referenced in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.heliyon.2024.e41323>.

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